

Self-distillation of context — experiment report

Project: self-distillation of context (same model as teacher and student).
Teacher sees privileged context (RAG passage / <think> trace); the student must reproduce its behavior without it. Corpus: 'La tierra de Alvargonzalez' (A. Machado, 1912), 725 verses, 228 tasks (continuations, per-section recitations, opening). Model: Qwen3-0.6B on a single RTX 3060 12 GB.

Methods: layer-wise hidden matching with block-local backward (summed / sequential / teacher_censored), plus local readout auxiliaries such as lens-CE and bounded tail-CE. Online teacher uses adapters-off as the frozen teacher, avoiding a disk cache.

Recitation (full corpus, n=333): base CER 0.897 ->
best (lw_k_tailonly_0p6b_rag) CER 0.008, 99% lines verbatim.
Forgetting probe (CE on held-out text): base 3.277,
best-run 4.609 (delta +1.33).
(computed from current artifacts)

Key findings (NARRATIVE SNAPSHOT written 2026-07-03 – tables and figure pages are computed from current artifacts; re-read them if this report was regenerated after new experiments):

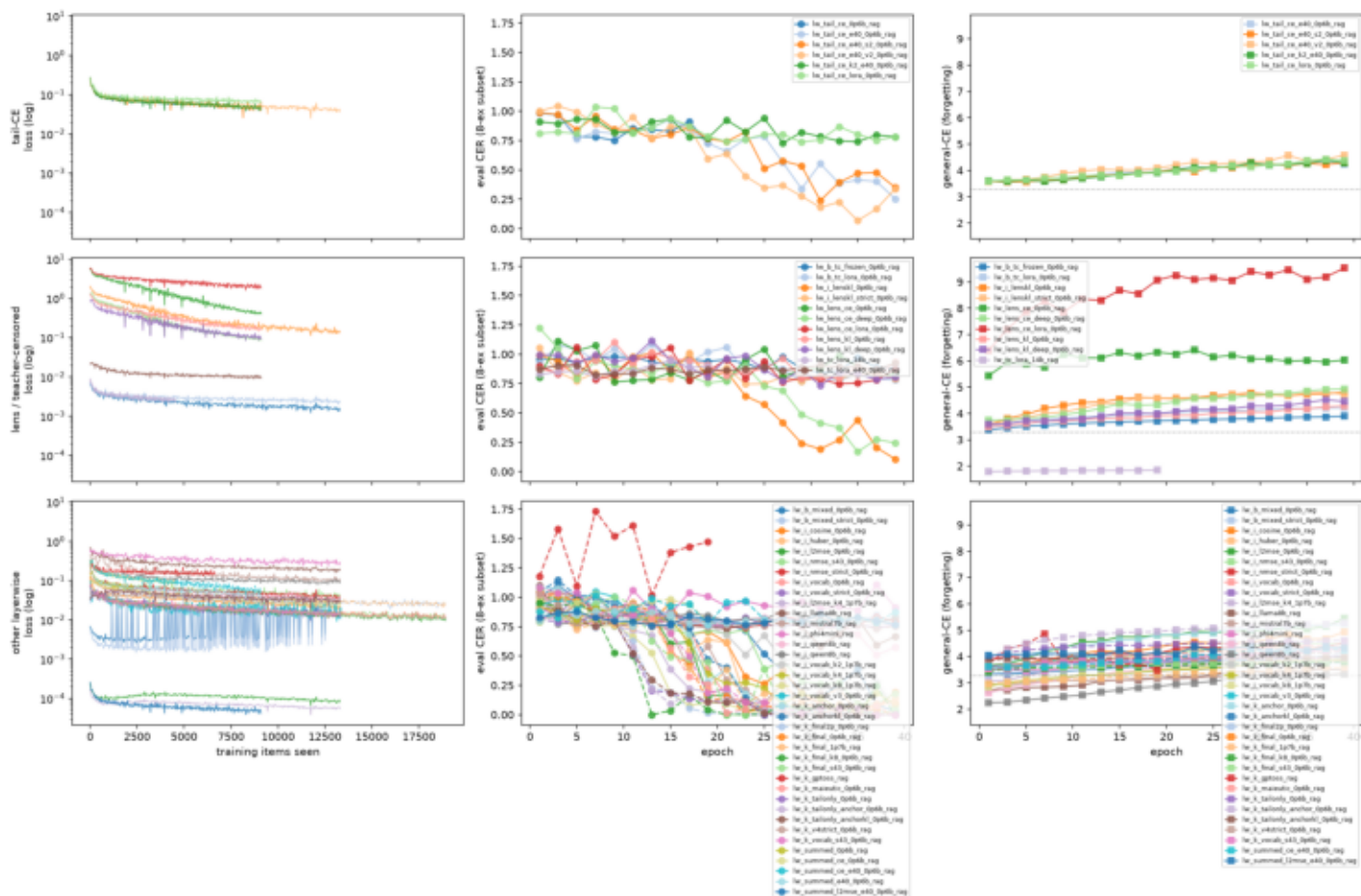
1. Strict hidden matching stores signal but does not reliably produce free-run behavior.
2. Tail-CE is the current best lever: it keeps hidden matching as the storage signal and spends bounded top-window credit on readout.
3. teacher_censored: per-layer increment targets are much smaller than student-stream targets, peak at layer ~7 (context integration site), ~0 at the last layer; layers train independently (parallelizable).
4. Sequential and teacher_censored schedules preserve the one-block memory story needed for very large models.

Generated 2026-07-04 07:22. Details in the following pages;
reproducibility: configs/experiments/*.yaml, runs/*/metrics.jsonl, git log.

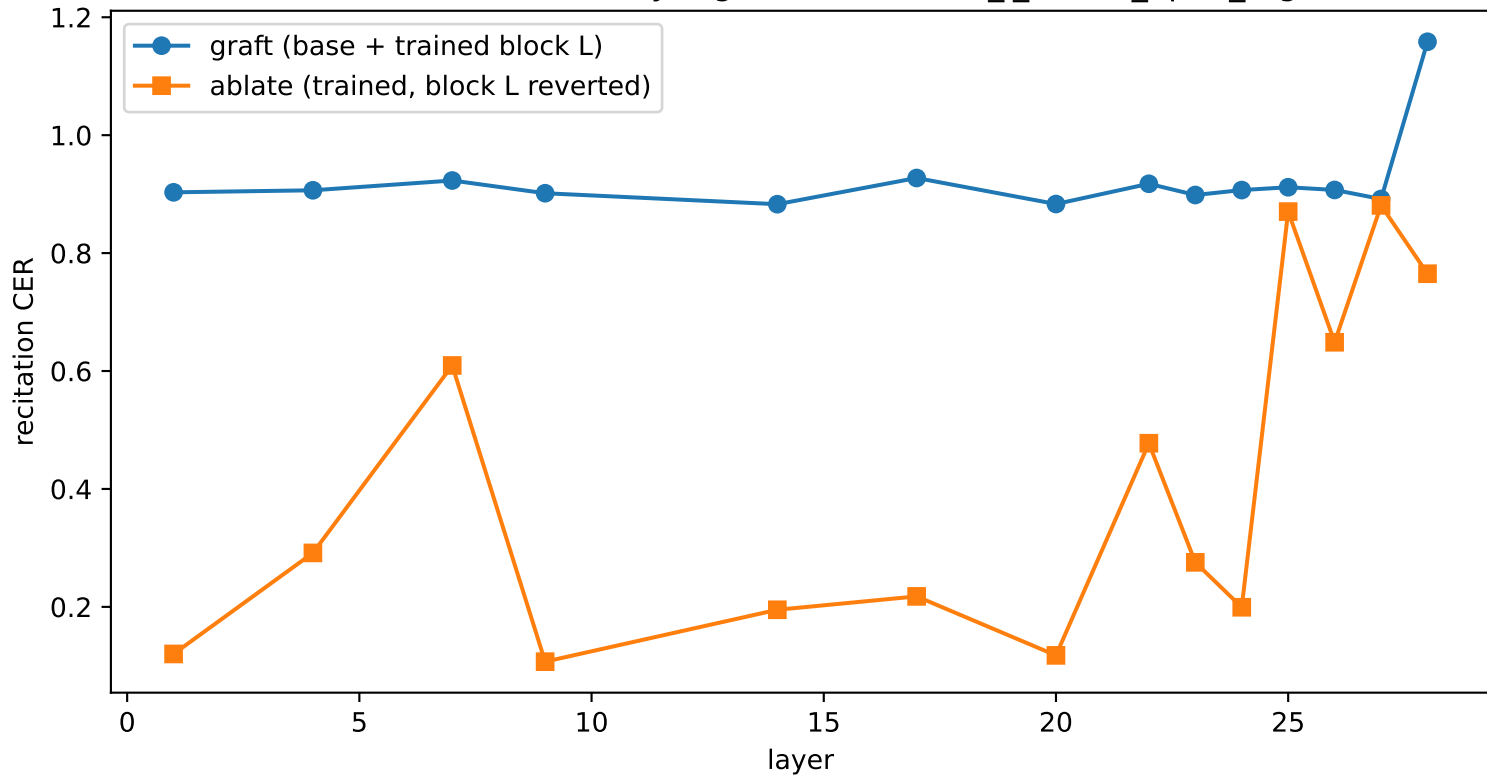
Results table (all runs)

run	method	sched	lora	mode	compact	train_cer	eval_cer	exact	forget	loss0	lossN	steps	items	vram_gb	min
lw_b_mixed_0p6b_rag	layerwise	mixed	False	rag	remove	0.0296	0.11	0.882	0.74	0.0066	0.017	1.66e+03	1.33e+04	9.92	51
lw_b_mixed_strict_0p6b_rag	layerwise	mixed	False	rag	remove	0.762	0.863	0	0.579	0.0054	0.0147	1.66e+03	1.33e+04	8.53	39.3
lw_b_tc_frozen_0p6b_rag	layerwise	teacher_censored	False	rag	remove	0.883	0.84	0	0.617	0.0066	0.0016	1.66e+03	1.33e+04	9.83	54.1
lw_b_tc_lora_0p6b_rag	layerwise	teacher_censored	True	rag	remove	0.78	0.868	0	0.929	0.0067	0.0024	1.66e+03	1.33e+04	2.42	51.6
lw_i_cosine_0p6b_rag	layerwise	summed	False	rag	remove	0.195	0.104	0.924	1.66	0.127	0.0221	1.66e+03	1.33e+04	8.63	25
lw_i_huber_0p6b_rag	layerwise	summed	False	rag	remove	0.139	0.0612	0.94	1.28	0.116	0.0217	1.66e+03	1.33e+04	8.64	25
lw_i_l2mse_0p6b_rag	layerwise	summed	False	rag	remove	0	0.0352	0.957	2.17	0.0002	0.0001	1.66e+03	1.33e+04	8.64	33.5
lw_i_lenskl_0p6b_rag	layerwise	summed	False	rag	remove	0.107	0.565	0.397	1.53	2.04	0.148	1.66e+03	1.33e+04	11.8	128
lw_i_lenskl_strict_0p6b_rag	layerwise	summed	False	rag	remove	0.792	0.811	0.0114	1.44	2.04	0.147	1.66e+03	1.33e+04	9.07	152
lw_i_nmse_s43_0p6b_rag	layerwise	summed	False	rag	remove	0.181	0.0921	0.898	1.21	0.26	0.0423	1.66e+03	1.33e+04	8.63	41.7
lw_i_nmse_strict_0p6b_rag	layerwise	summed	False	rag	remove	0.796	0.851	0	1.26	0.261	0.0401	1.66e+03	1.33e+04	7.25	24.1
lw_i_vocab_0p6b_rag	layerwise	summed	False	rag	remove	0.0994	0.0241	0.978	1.01	0.076	0.014	1.66e+03	1.33e+04	8.64	43.8
lw_i_vocab_strict_0p6b_rag	layerwise	summed	False	rag	remove	0.76	0.849	0	0.724	0.0759	0.0138	1.66e+03	1.33e+04	7.25	25
lw_j_l2mse_k4_1p7b_rag	layerwise	summed	False	rag	remove	0	0.0121	0.985	1.48	0.0002	0.0001	1.66e+03	1.33e+04	24.4	36.6
lw_j_llama8b_rag	layerwise	summed	True	rag	remove	0.665	0.26	0.664	—	0.683	0.183	1.66e+03	1.33e+04	17.3	122
lw_j_mistral7b_rag	layerwise	summed	True	rag	remove	0	0.0536	0.927	—	0.441	0.104	1.66e+03	1.33e+04	15.4	95.4
lw_j_phi4mini_rag	layerwise	summed	True	rag	remove	0.914	0.918	0	—	0.626	0.294	1.66e+03	1.33e+04	8.87	67
lw_j_qwen4b_rag	layerwise	summed	True	rag	remove	0.574	0.244	0.729	—	0.186	0.0331	1.66e+03	1.33e+04	9.6	94.8
lw_j_qwen8b_rag	layerwise	summed	True	rag	remove	0.83	0.419	0.487	—	0.365	0.089	1.66e+03	1.33e+04	17.9	135
lw_j_vocab_k2_1p7b_rag	layerwise	summed	False	rag	remove	0.0417	0.0867	0.907	1.16	0.161	0.0281	1.66e+03	1.33e+04	24	31.9
lw_j_vocab_k4_1p7b_rag	layerwise	summed	False	rag	remove	0.0898	0.0754	0.923	1.17	0.162	0.0278	1.66e+03	1.33e+04	24.4	70.5
lw_j_vocab_k8_1p7b_rag	layerwise	summed	False	rag	remove	0	0.0415	0.959	1.19	0.162	0.0236	1.66e+03	1.33e+04	25.1	31.9
lw_j_vocab_v3_0p6b_rag	layerwise	summed	False	rag	remove	0.879	0.321	0.601	0.744	0.0587	0.0172	1.67e+03	1.34e+04	8.66	45.6
lw_k_anchor_0p6b_rag	layerwise	summed	False	rag	remove	0	0.023	0.97	1.93	0.076	0.0146	1.66e+03	1.33e+04	8.66	25.2
lw_k_anchorkl_0p6b_rag	layerwise	summed	False	rag	remove	0	0.0212	0.976	0.498	0.0761	0.0143	1.66e+03	1.33e+04	9.92	30
lw_k_final2p_0p6b_rag	layerwise	tail_only	False	rag	remove	0	0.00896	0.988	0.853	0.031	0.026	2.37e+03	1.9e+04	4.41	18.3
lw_k_final_0p6b_rag	layerwise	summed	False	rag	remove	0	0.00921	0.987	0.586	0.0738	0.0125	2.37e+03	1.9e+04	9.92	36.3
lw_k_final_1p7b_rag	layerwise	summed	False	rag	remove	0.101	0.0205	0.976	0.87	0.16	0.0242	2.37e+03	1.9e+04	27.9	53.7
lw_k_final_k8_0p6b_rag	layerwise	summed	False	rag	remove	0	0.0151	0.986	0.514	0.0736	0.0105	2.37e+03	1.9e+04	10.2	32.7
lw_k_final_s43_0p6b_rag	layerwise	summed	False	rag	remove	0.0417	0.0124	0.988	0.641	0.0754	0.0119	2.37e+03	1.9e+04	9.92	31.9
lw_k_gptoss_rag	layerwise	summed	True	rag	remove	1.47	1	0	—	0.273	0.148	832	6.66e+03	40.7	33.7

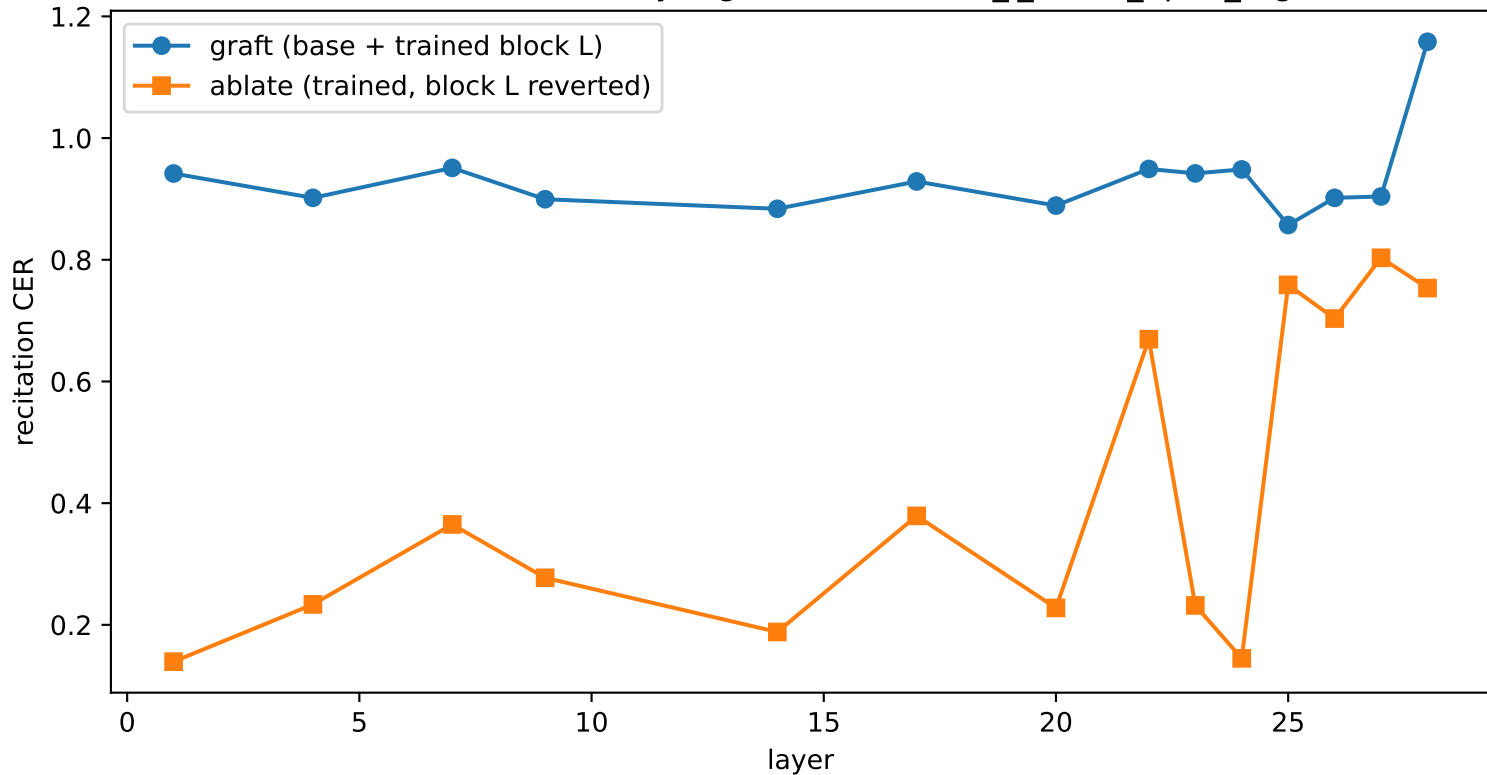
Training dynamics (loss / eval CER)



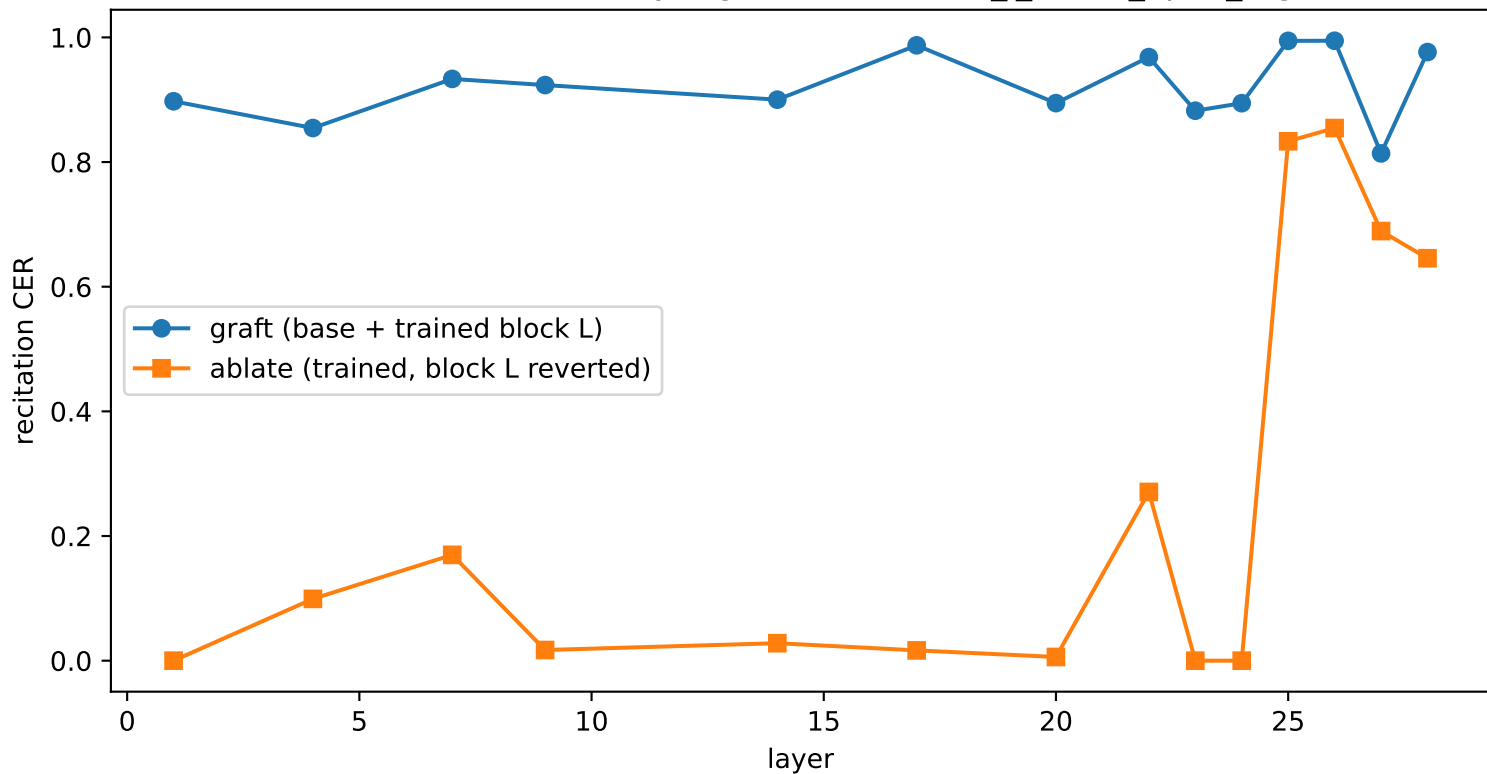
Causal localization: layer graft/ablate — lw_i_cosine_0p6b_rag



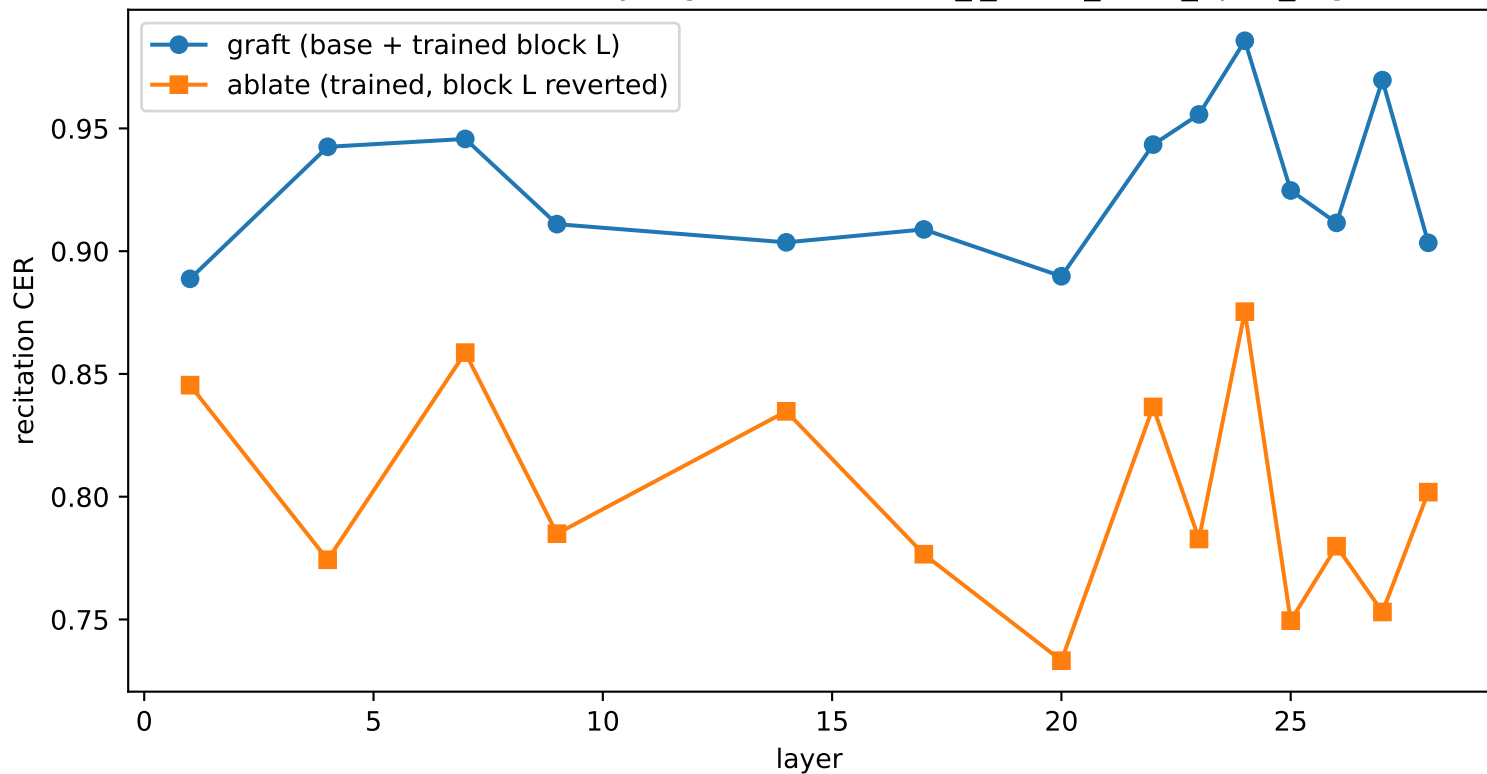
Causal localization: layer graft/ablate — lw_i_huber_0p6b_rag



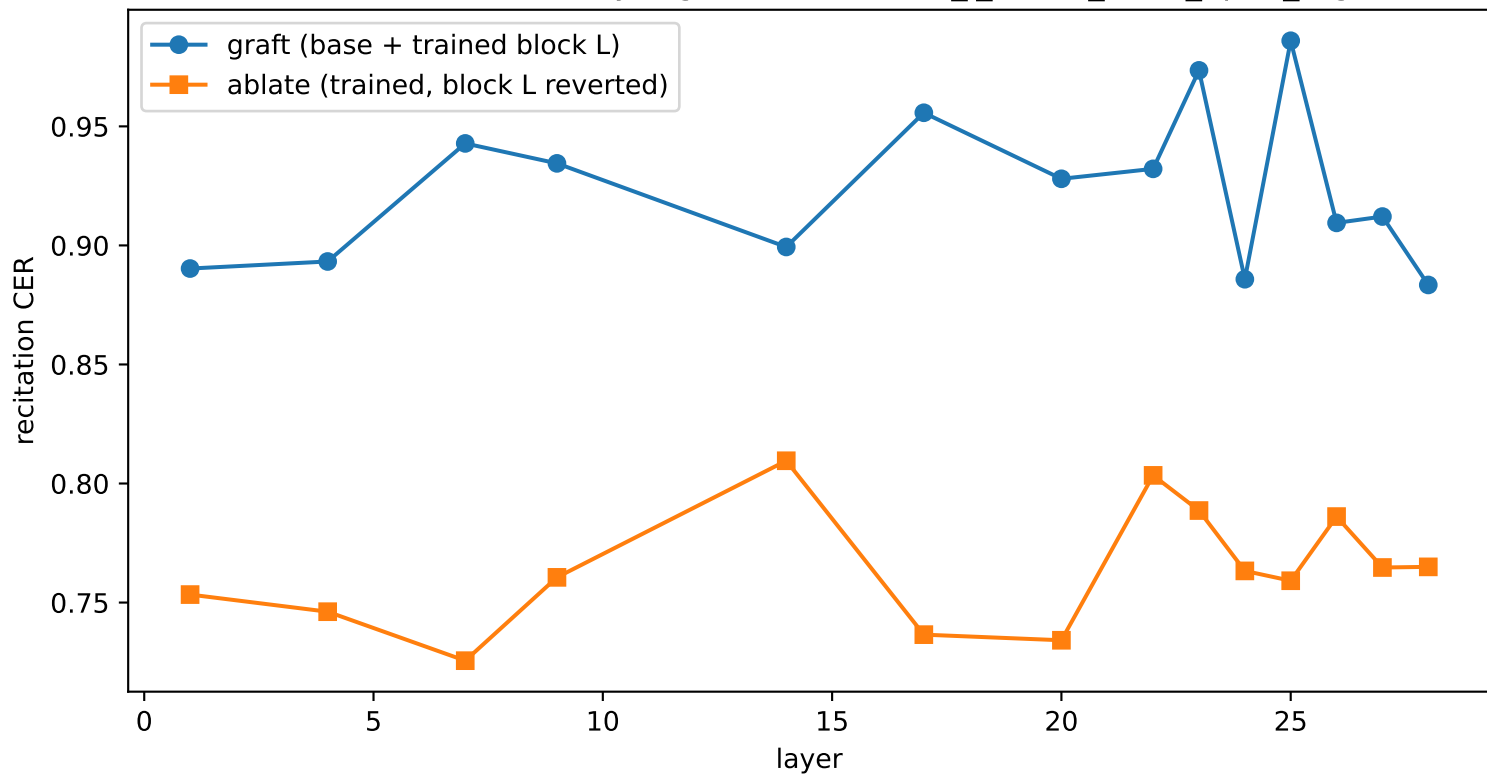
Causal localization: layer graft/ablate — lw_i_l2mse_0p6b_rag



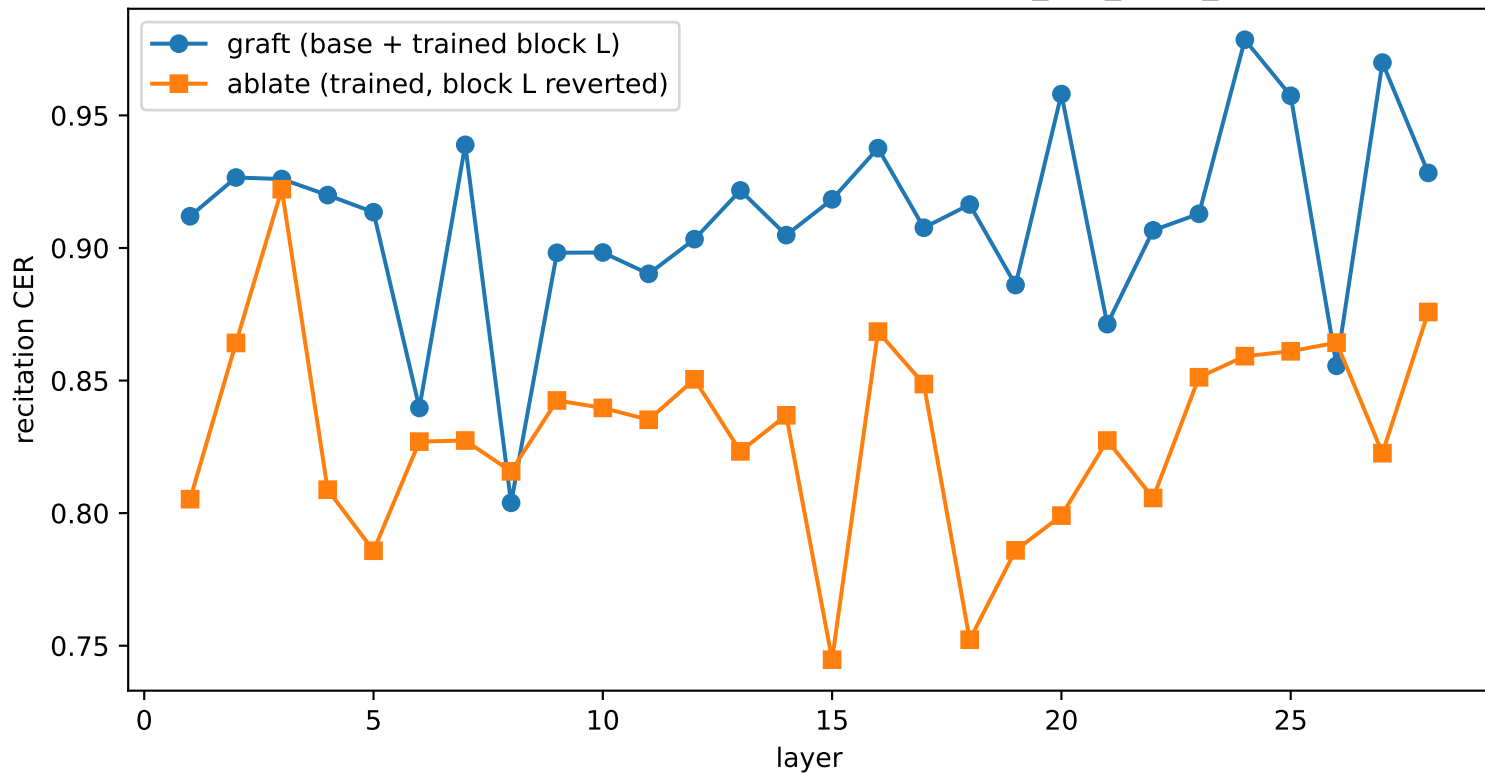
Causal localization: layer graft/ablate — lw_i_nmse_strict_0p6b_rag



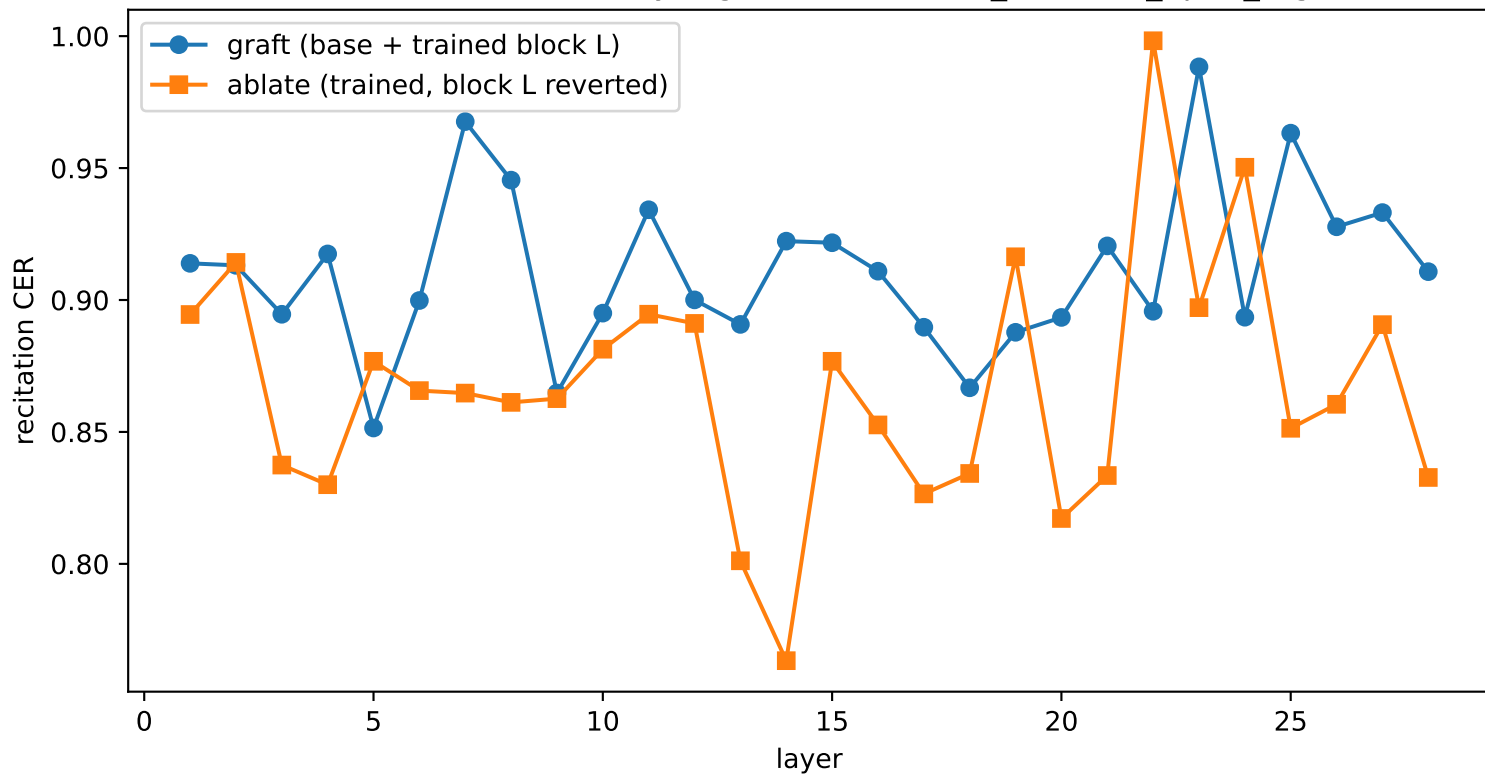
Causal localization: layer graft/ablate — lw_i_vocab_strict_0p6b_rag



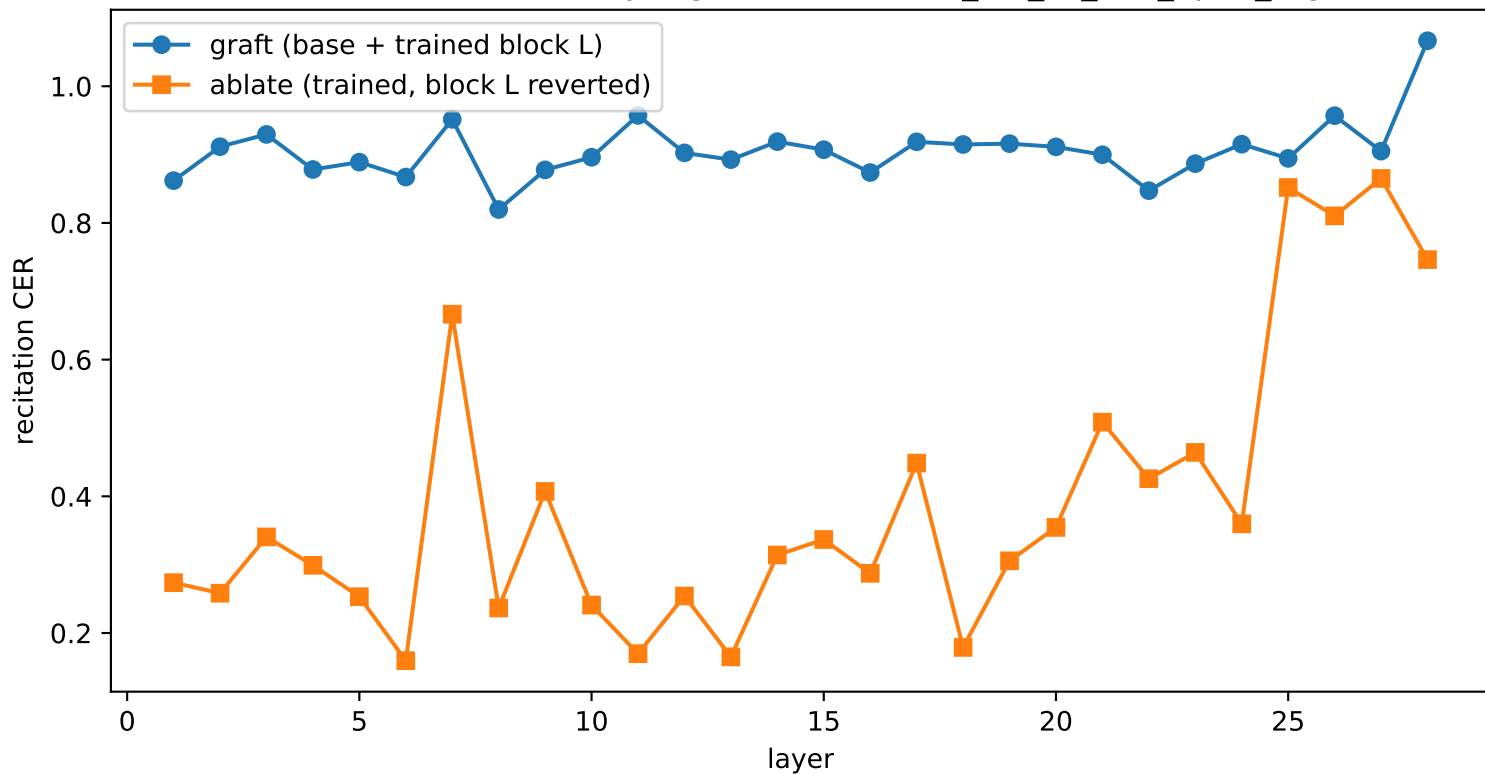
Causal localization: layer graft/ablate — lw_seq_0p6b_rag



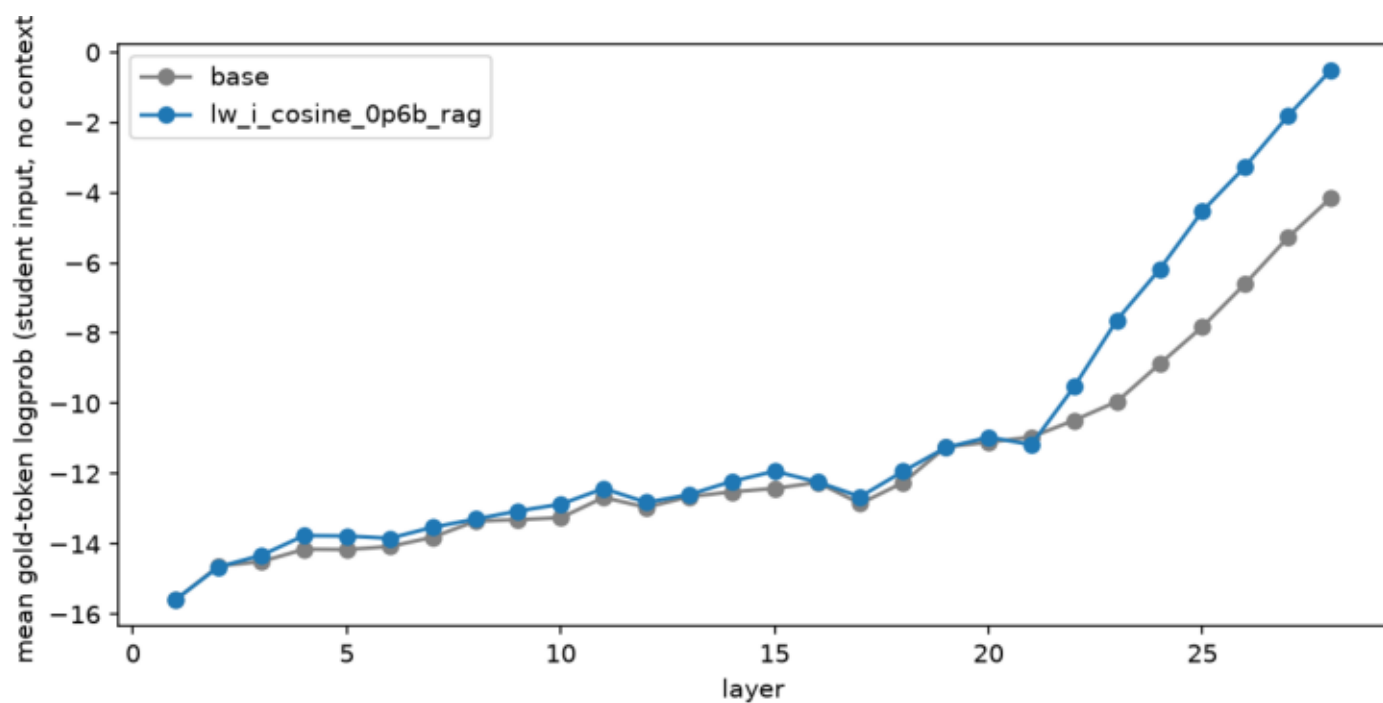
Causal localization: layer graft/ablate — lw_summed_0p6b_rag



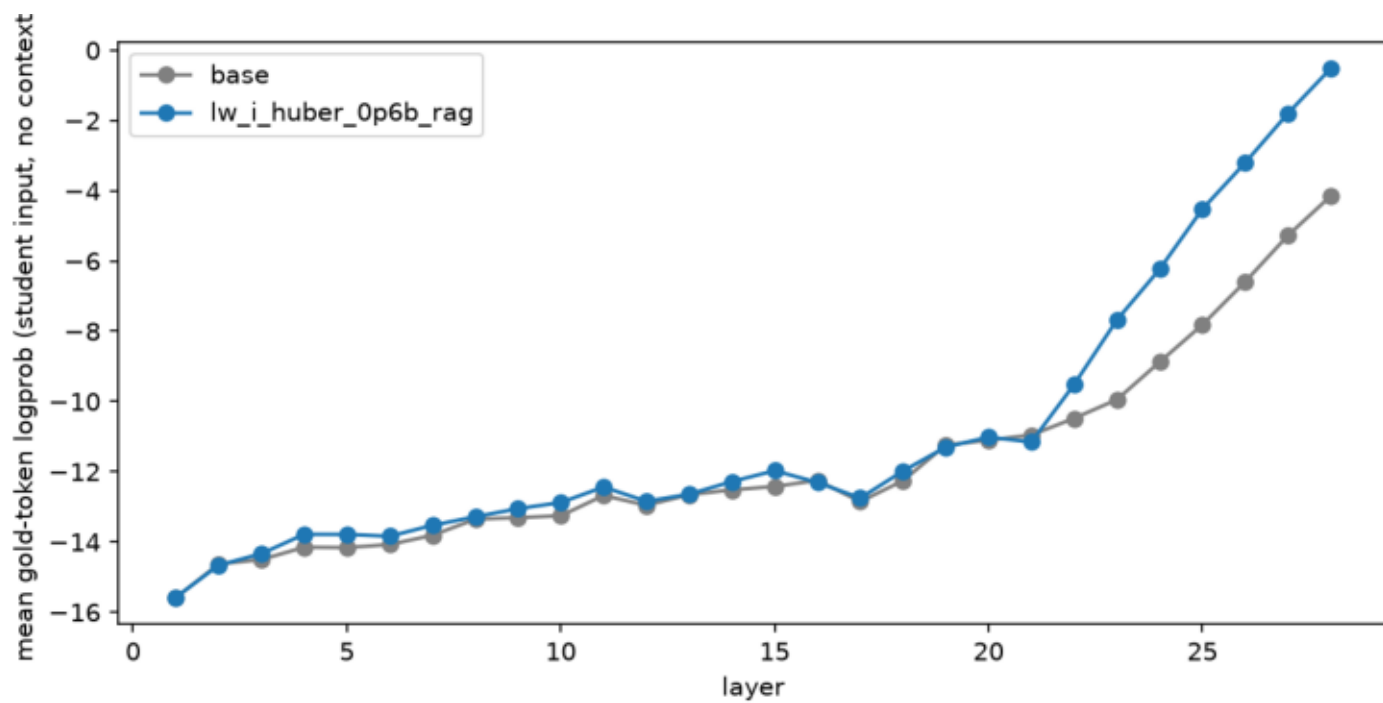
Causal localization: layer graft/ablate — lw_tail_ce_e40_0p6b_rag



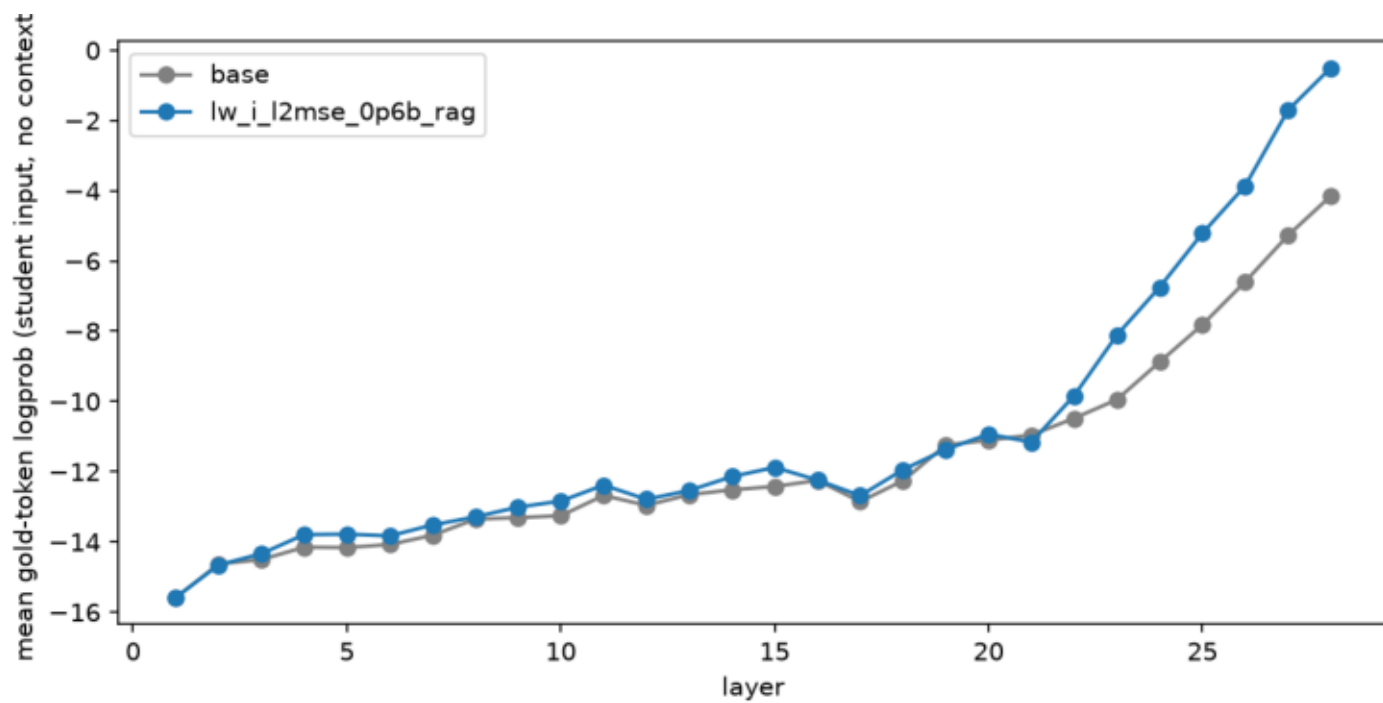
Logit-lens depth profile — lw_i_cosine_0p6b_rag



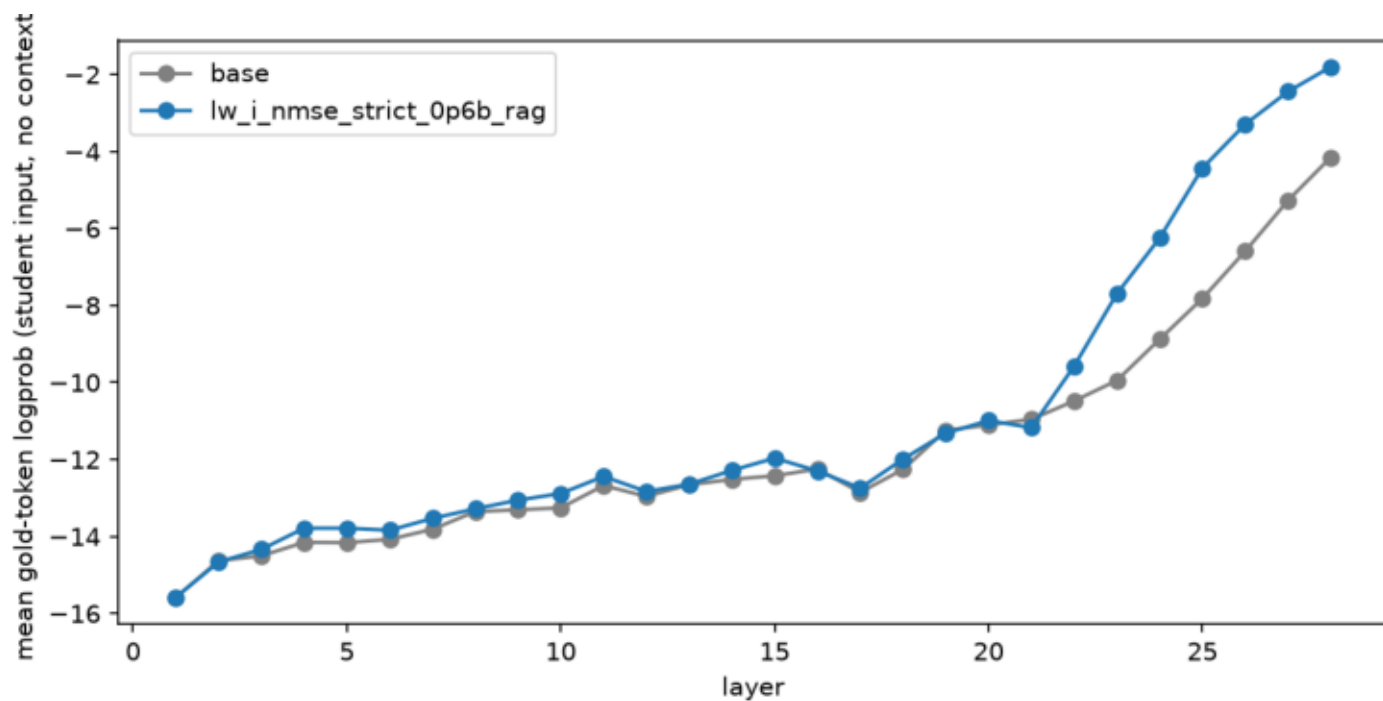
Logit-lens depth profile — lw_i_huber_0p6b_rag



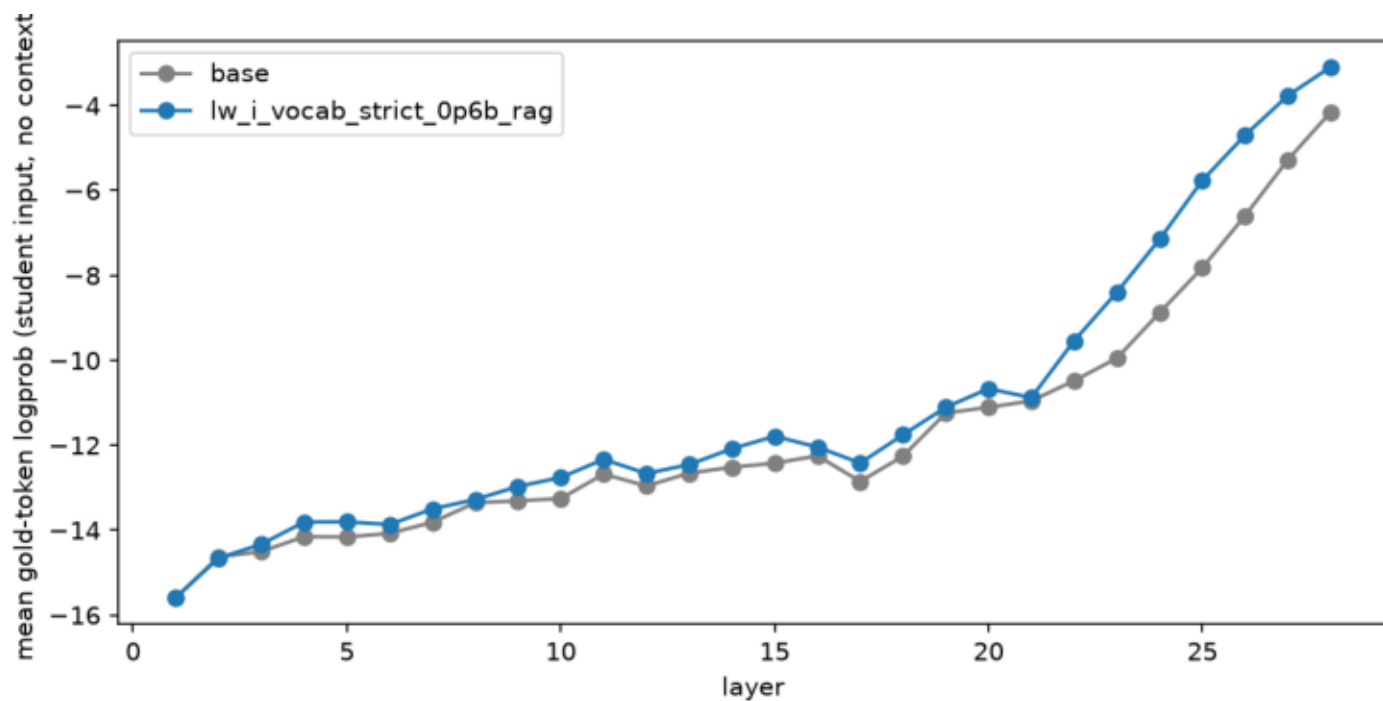
Logit-lens depth profile — lw_i_l2mse_0p6b_rag



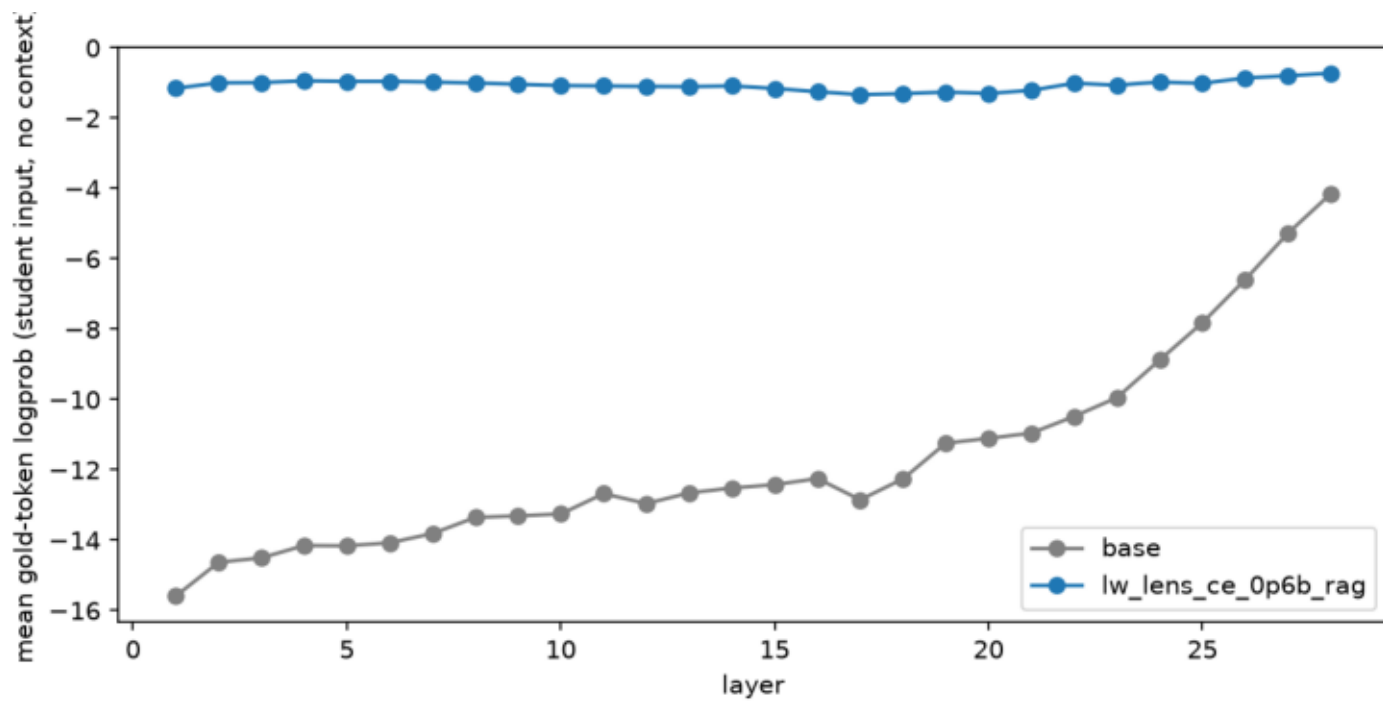
Logit-lens depth profile — lw_i_nmse_strict_0p6b_rag



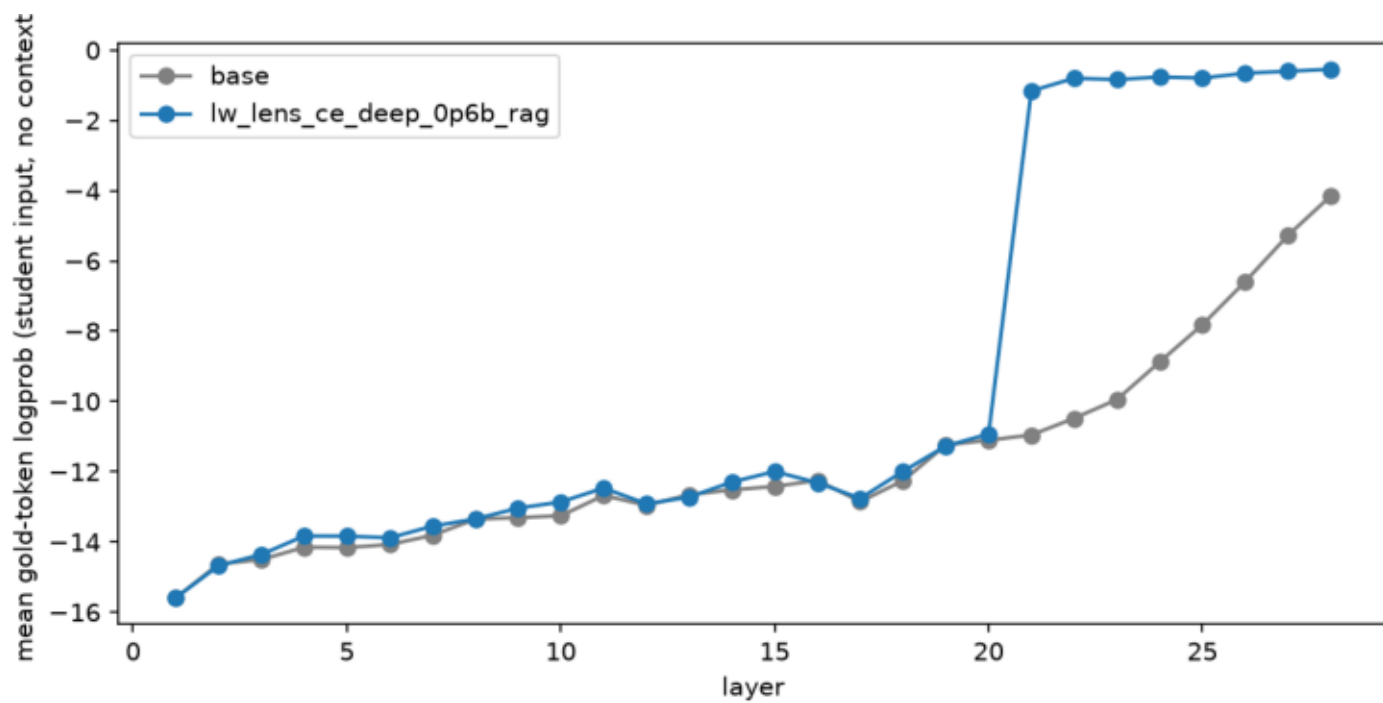
Logit-lens depth profile — lw_i_vocab_strict_0p6b_rag



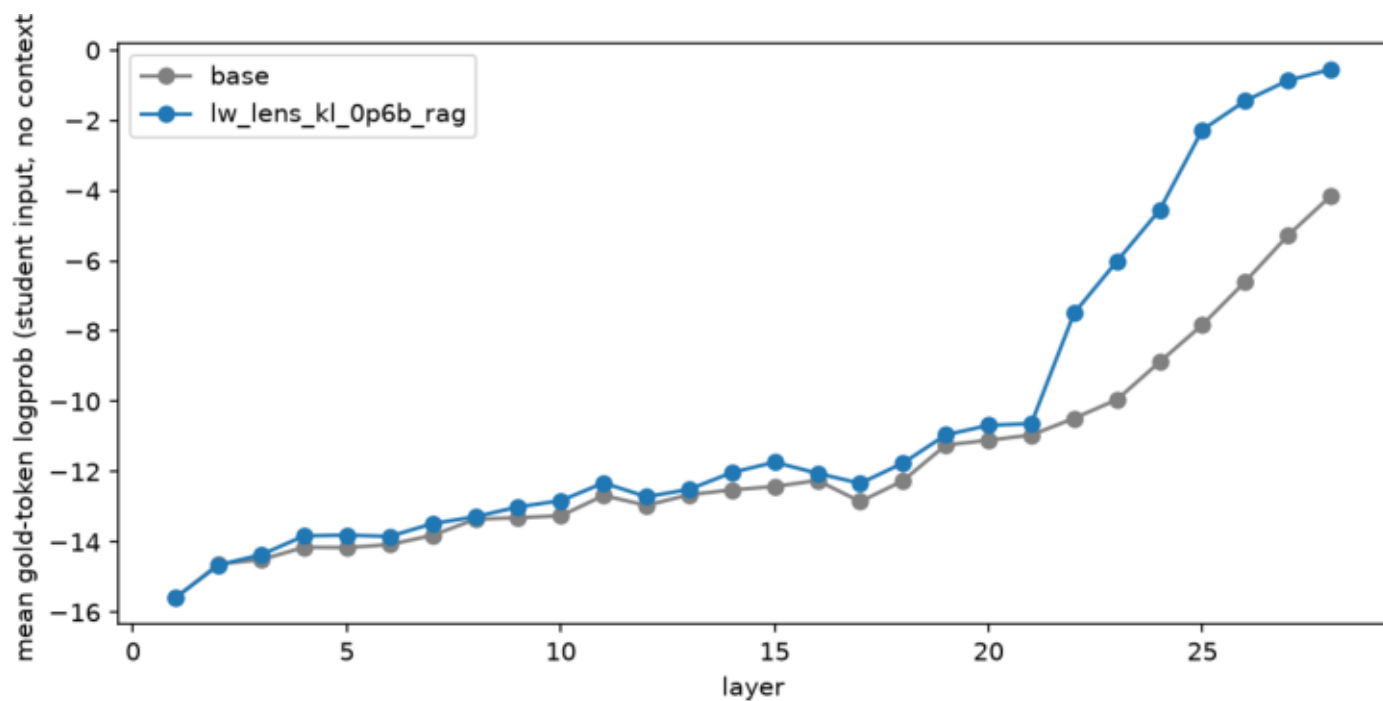
Logit-lens depth profile — lw_lens_ce_0p6b_rag



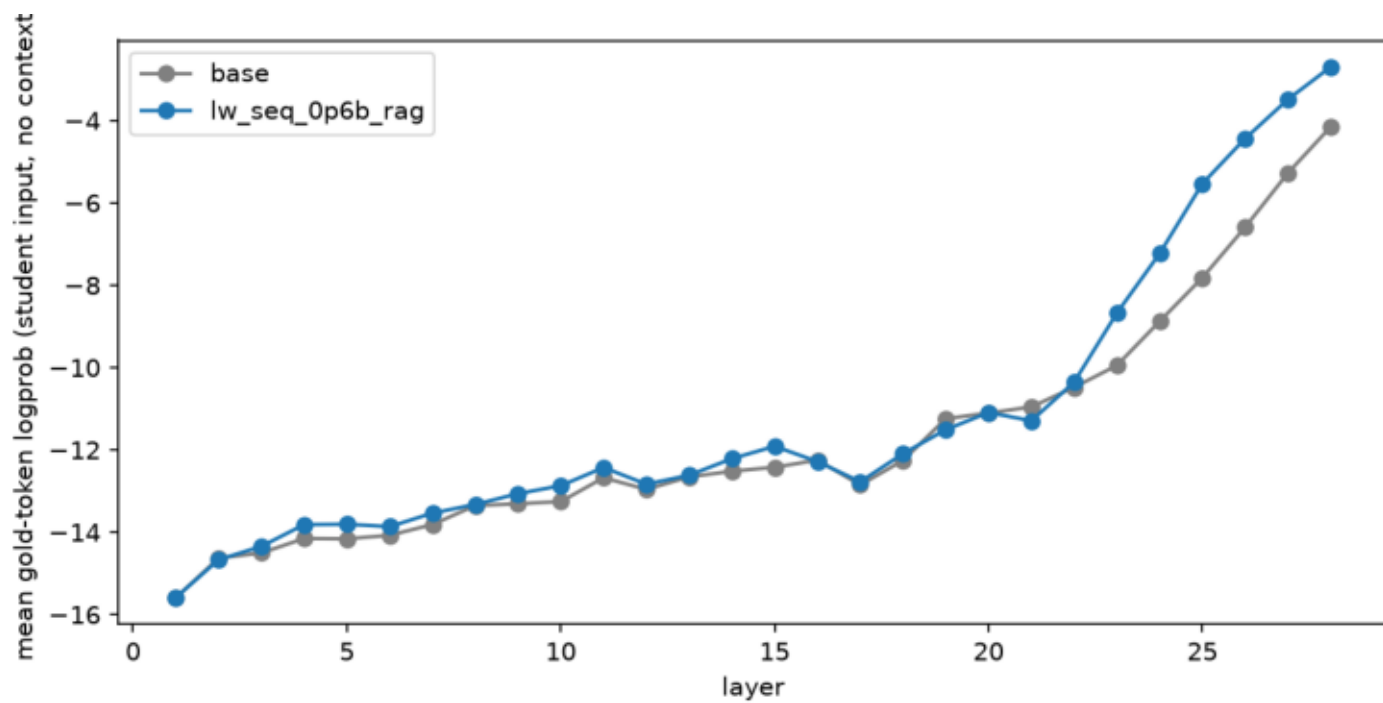
Logit-lens depth profile — lw_lens_ce_deep_0p6b_rag



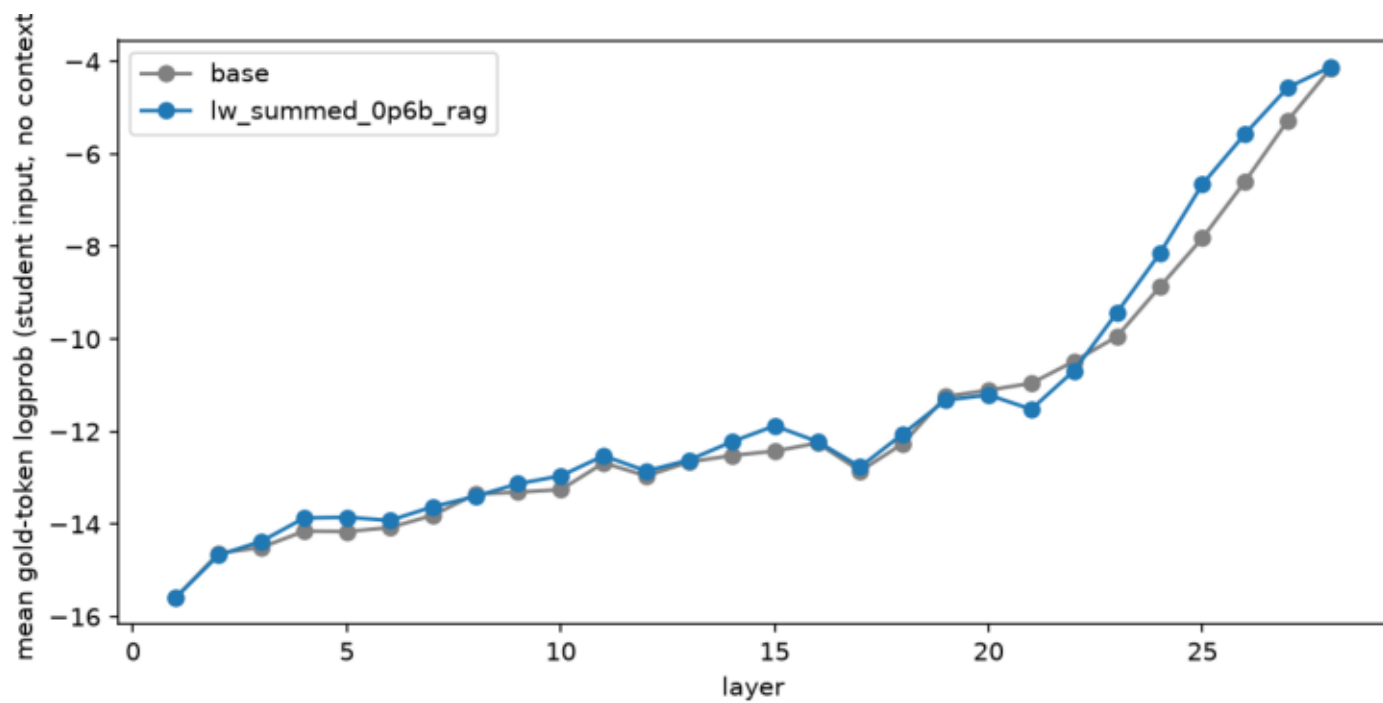
Logit-lens depth profile — lw_lens_kl_0p6b_rag



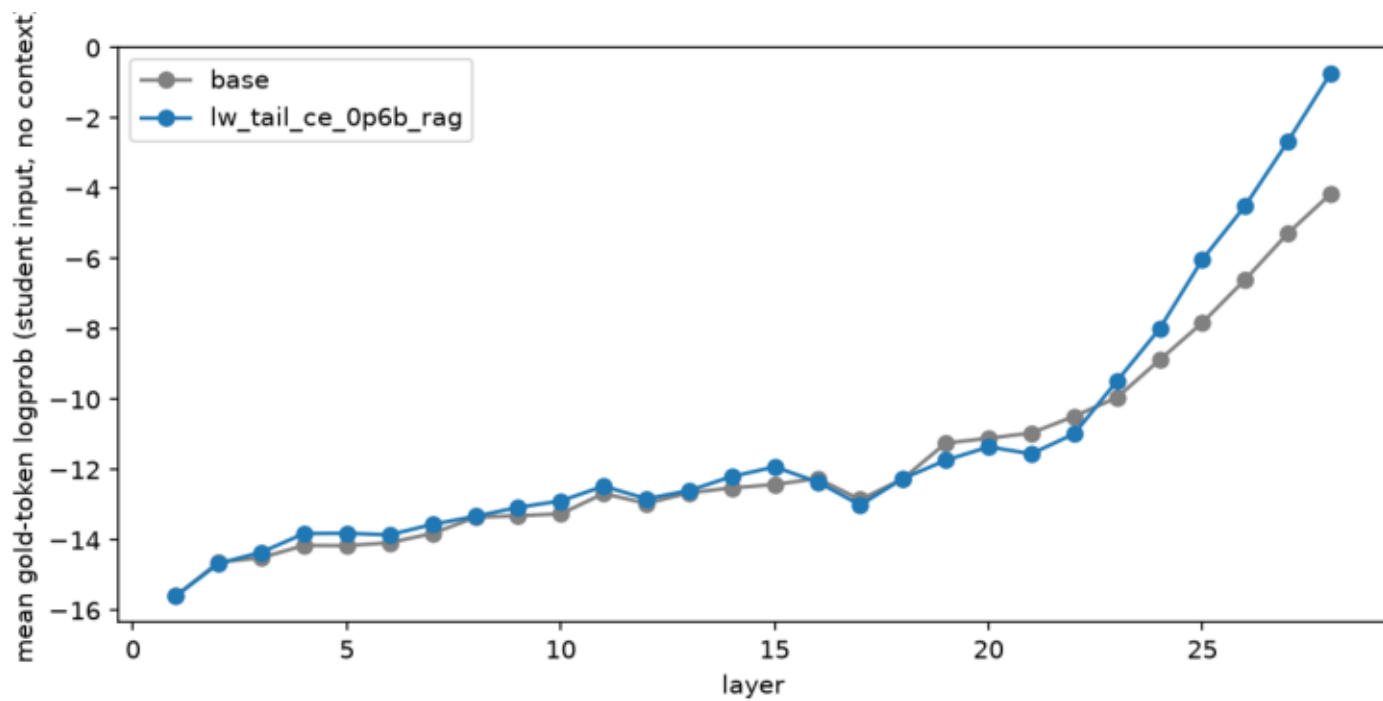
Logit-lens depth profile — lw_seq_0p6b_rag



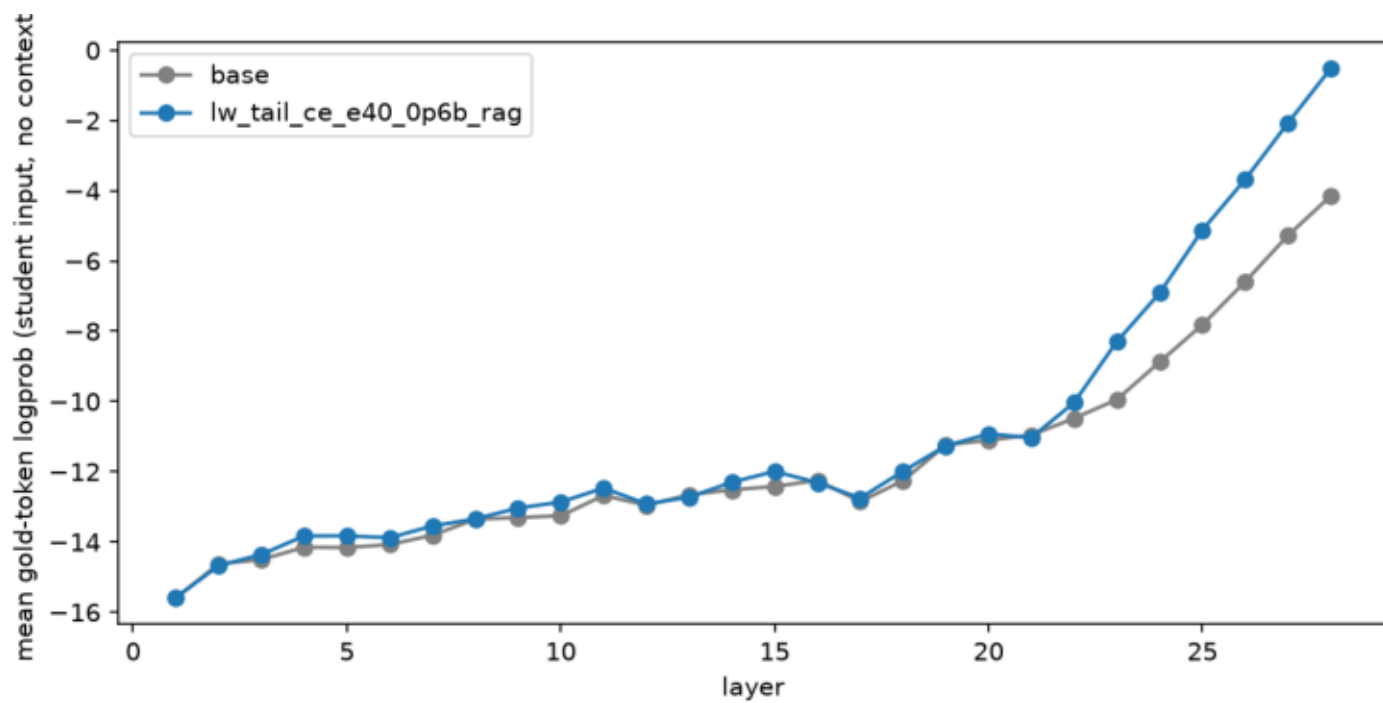
Logit-lens depth profile — lw_summed_0p6b_rag



Logit-lens depth profile — lw_tail_ce_0p6b_rag



Logit-lens depth profile — lw_tail_ce_e40_0p6b_rag



Per-run details (1)

```
== lw_b_mixed_0p6b_rag ==
method=layerwise schedule=mixed lora=False lr=1e-05 epochs=40 ce=0.0/0.0 online=False
loss: first20 0.0066 -> last20 0.0170 (13320 items)
eval ep37: CER 0.086 exact 0.89 vram 9.92GB 48.4min
eval ep39: CER 0.030 exact 1.00 vram 9.92GB 51.0min
FULL eval: CER 0.1102 line-exact 0.8820 general-CE 4.016

== lw_b_mixed_strict_0p6b_rag ==
method=layerwise schedule=mixed lora=False lr=1e-05 epochs=40 ce=0.0/0.0 online=False
loss: first20 0.0054 -> last20 0.0147 (13320 items)
eval ep37: CER 0.854 exact 0.00 vram 8.53GB 37.4min
eval ep39: CER 0.762 exact 0.00 vram 8.53GB 39.3min
FULL eval: CER 0.8632 line-exact 0.0000 general-CE 3.856

== lw_b_tc_frozen_0p6b_rag ==
method=layerwise schedule=teacher censored lora=False lr=1e-05 epochs=40 ce=0.0/0.0 online=False
loss: first20 0.0066 -> last20 0.0016 (13320 items)
eval ep37: CER 0.887 exact 0.00 vram 9.83GB 51.3min
eval ep39: CER 0.883 exact 0.00 vram 9.83GB 54.1min
FULL eval: CER 0.8397 line-exact 0.0000 general-CE 3.894

== lw_b_tc_lora_0p6b_rag ==
method=layerwise schedule=teacher censored lora=True lr=0.0001 epochs=40 ce=0.0/0.0 online=True
loss: first20 0.0067 -> last20 0.0024 (13320 items)
eval ep37: CER 0.795 exact 0.00 vram 2.42GB 49.6min
eval ep39: CER 0.780 exact 0.00 vram 2.42GB 51.6min
FULL eval: CER 0.8683 line-exact 0.0000 general-CE 4.206

== lw_i_cosine_0p6b_rag ==
method=layerwise schedule=summed lora=False lr=1e-05 epochs=40 ce=0.0/0.0 online=False
loss: first20 0.1273 -> last20 0.0221 (13320 items)
eval ep37: CER 0.056 exact 0.95 vram 8.63GB 23.7min
eval ep39: CER 0.195 exact 1.00 vram 8.63GB 25.0min
FULL eval: CER 0.1037 line-exact 0.9238 general-CE 4.939

== lw_i_huber_0p6b_rag ==
method=layerwise schedule=summed lora=False lr=1e-05 epochs=40 ce=0.0/0.0 online=False
loss: first20 0.1163 -> last20 0.0217 (13320 items)
eval ep37: CER 0.056 exact 0.95 vram 8.64GB 23.7min
eval ep39: CER 0.139 exact 0.88 vram 8.64GB 25.0min
FULL eval: CER 0.0612 line-exact 0.9399 general-CE 4.558
```


Per-run details (2)

```
== lw_i_l2mse_0p6b_rag ==
method=layerwise schedule=summed lora=False lr=1e-05 epochs=40 ce=0.0/0.0 online=False
loss: first20 0.0002 -> last20 0.0001 (13320 items)
eval ep37: CER 0.000 exact 1.00 vram 8.64GB 31.2min
eval ep39: CER 0.000 exact 1.00 vram 8.64GB 33.5min
FULL eval: CER 0.0352 line-exact 0.9572 general-CE 5.450

== lw_i_lenskl_0p6b_rag ==
method=layerwise schedule=summed lora=False lr=1e-05 epochs=40 ce=0.0/0.0 online=False
loss: first20 2.0438 -> last20 0.1476 (13320 items)
eval ep37: CER 0.205 exact 0.53 vram 11.84GB 119.9min
eval ep39: CER 0.107 exact 0.84 vram 11.84GB 128.4min
FULL eval: CER 0.5655 line-exact 0.3966 general-CE 4.808

== lw_i_lenskl_strict_0p6b_rag ==
method=layerwise schedule=summed lora=False lr=1e-05 epochs=40 ce=0.0/0.0 online=False
loss: first20 2.0437 -> last20 0.1466 (13320 items)
eval ep37: CER 0.838 exact 0.00 vram 9.07GB 143.2min
eval ep39: CER 0.792 exact 0.00 vram 9.07GB 152.3min
FULL eval: CER 0.8114 line-exact 0.0114 general-CE 4.714

== lw_i_nmse_s43_0p6b_rag ==
method=layerwise schedule=summed lora=False lr=1e-05 epochs=40 ce=0.0/0.0 online=False
loss: first20 0.2600 -> last20 0.0423 (13320 items)
eval ep37: CER 0.093 exact 0.91 vram 8.63GB 39.5min
eval ep39: CER 0.181 exact 0.75 vram 8.63GB 41.7min
FULL eval: CER 0.0921 line-exact 0.8982 general-CE 4.484

== lw_i_nmse_strict_0p6b_rag ==
method=layerwise schedule=summed lora=False lr=1e-05 epochs=40 ce=0.0/0.0 online=False
loss: first20 0.2614 -> last20 0.0401 (13320 items)
eval ep37: CER 0.832 exact 0.00 vram 7.25GB 23.0min
eval ep39: CER 0.796 exact 0.00 vram 7.25GB 24.1min
FULL eval: CER 0.8505 line-exact 0.0000 general-CE 4.535

== lw_i_vocab_0p6b_rag ==
method=layerwise schedule=summed lora=False lr=1e-05 epochs=40 ce=0.0/0.0 online=False
loss: first20 0.0760 -> last20 0.0140 (13320 items)
eval ep37: CER 0.092 exact 0.88 vram 8.64GB 41.0min
eval ep39: CER 0.099 exact 1.00 vram 8.64GB 43.8min
FULL eval: CER 0.0241 line-exact 0.9783 general-CE 4.283
```

Per-run details (3)

```
== lw_i_vocab_strict_0p6b_rag ==
method=layerwise schedule=summed lora=False lr=1e-05 epochs=40 ce=0.0/0.0 online=False
loss: first20 0.0759 -> last20 0.0138 (13320 items)
eval ep37: CER 0.744 exact 0.00 vram 7.25GB 24.0min
eval ep39: CER 0.760 exact 0.00 vram 7.25GB 25.0min
FULL eval: CER 0.8490 line-exact 0.0000 general-CE 4.000

== lw_j_l2mse_k4_1p7b_rag ==
method=layerwise schedule=summed lora=False lr=1e-05 epochs=40 ce=0.0/0.0 online=False
loss: first20 0.0002 -> last20 0.0001 (13320 items)
eval ep37: CER 0.000 exact 1.00 vram 24.38GB 35.0min
eval ep39: CER 0.000 exact 1.00 vram 24.38GB 36.6min
FULL eval: CER 0.0121 line-exact 0.9845 general-CE 4.215

== lw_j_llama8b_rag ==
method=layerwise schedule=summed lora=True lr=0.0001 epochs=40 ce=0.0/0.0 online=True
loss: first20 0.6829 -> last20 0.1827 (13320 items)
eval ep37: CER 0.593 exact 0.24 vram 17.34GB 115.2min
eval ep39: CER 0.665 exact 0.12 vram 17.34GB 122.2min
FULL eval: CER 0.2597 line-exact 0.6639 general-CE 3.481

== lw_j_mistral7b_rag ==
method=layerwise schedule=summed lora=True lr=0.0001 epochs=40 ce=0.0/0.0 online=True
loss: first20 0.4409 -> last20 0.1043 (13320 items)
eval ep37: CER 0.000 exact 1.00 vram 15.44GB 91.9min
eval ep39: CER 0.000 exact 1.00 vram 15.44GB 95.4min
FULL eval: CER 0.0536 line-exact 0.9273 general-CE 3.375

== lw_j_phi4mini_rag ==
method=layerwise schedule=summed lora=True lr=0.0001 epochs=40 ce=0.0/0.0 online=True
loss: first20 0.6259 -> last20 0.2945 (13320 items)
eval ep37: CER 1.106 exact 0.00 vram 8.87GB 64.0min
eval ep39: CER 0.914 exact 0.00 vram 8.87GB 67.0min
FULL eval: CER 0.9183 line-exact 0.0000 general-CE 4.277

== lw_j_qwen4b_rag ==
method=layerwise schedule=summed lora=True lr=0.0001 epochs=40 ce=0.0/0.0 online=True
loss: first20 0.1856 -> last20 0.0331 (13320 items)
eval ep37: CER 0.508 exact 0.38 vram 9.6GB 90.4min
eval ep39: CER 0.574 exact 0.25 vram 9.6GB 94.8min
FULL eval: CER 0.2439 line-exact 0.7286 general-CE 3.913
```

Per-run details (4)

```
== lw_j_qwen8b_rag ==
method=layerwise schedule=summed lora=True lr=0.0001 epochs=40 ce=0.0/0.0 online=True
loss: first20 0.3654 -> last20 0.0890 (13320 items)
eval ep37: CER 0.773 exact 0.05 vram 17.85GB 128.3min
eval ep39: CER 0.830 exact 0.00 vram 17.85GB 135.3min
FULL eval: CER 0.4185 line-exact 0.4870 general-CE 3.320

== lw_j_vocab_k2_1p7b_rag ==
method=layerwise schedule=summed lora=False lr=1e-05 epochs=40 ce=0.0/0.0 online=False
loss: first20 0.1614 -> last20 0.0281 (13320 items)
eval ep37: CER 0.276 exact 0.62 vram 24.04GB 30.3min
eval ep39: CER 0.042 exact 1.00 vram 24.04GB 31.9min
FULL eval: CER 0.0867 line-exact 0.9073 general-CE 3.887

== lw_j_vocab_k4_1p7b_rag ==
method=layerwise schedule=summed lora=False lr=1e-05 epochs=40 ce=0.0/0.0 online=False
loss: first20 0.1615 -> last20 0.0278 (13320 items)
eval ep37: CER 0.186 exact 0.75 vram 24.38GB 68.9min
eval ep39: CER 0.090 exact 0.88 vram 24.38GB 70.5min
FULL eval: CER 0.0754 line-exact 0.9229 general-CE 3.903

== lw_j_vocab_k8_1p7b_rag ==
method=layerwise schedule=summed lora=False lr=1e-05 epochs=40 ce=0.0/0.0 online=False
loss: first20 0.1616 -> last20 0.0236 (13320 items)
eval ep37: CER 0.000 exact 1.00 vram 25.05GB 30.3min
eval ep39: CER 0.000 exact 1.00 vram 25.05GB 31.9min
FULL eval: CER 0.0415 line-exact 0.9591 general-CE 3.922

== lw_j_vocab_v3_0p6b_rag ==
method=layerwise schedule=summed lora=False lr=1e-05 epochs=14 ce=0.0/0.0 online=False
loss: first20 0.0587 -> last20 0.0172 (13398 items)
eval ep11: CER 0.834 exact 0.00 vram 8.66GB 39.4min
eval ep13: CER 0.879 exact 0.00 vram 8.66GB 45.6min
FULL eval: CER 0.3212 line-exact 0.6013 general-CE 4.020

== lw_k_anchor_0p6b_rag ==
method=layerwise schedule=summed lora=False lr=1e-05 epochs=40 ce=0.0/0.0 online=False
loss: first20 0.0760 -> last20 0.0146 (13320 items)
eval ep37: CER 0.042 exact 1.00 vram 8.66GB 23.9min
eval ep39: CER 0.000 exact 1.00 vram 8.66GB 25.2min
FULL eval: CER 0.0230 line-exact 0.9704 general-CE 5.209
```

Per-run details (5)

```
== lw_k_anchorkl_0p6b_rag ==
method=layerwise schedule=summed lora=False lr=1e-05 epochs=40 ce=0.0/0.0 online=False
loss: first20 0.0761 -> last20 0.0143 (13320 items)
eval ep37: CER 0.000 exact 1.00 vram 9.92GB 28.8min
eval ep39: CER 0.000 exact 1.00 vram 9.92GB 30.0min
FULL eval: CER 0.0212 line-exact 0.9760 general-CE 3.775

== lw_k_final2p_0p6b_rag ==
method=layerwise schedule=tail_only lora=False lr=1e-05 epochs=40 ce=0.0/0.0 online=False
loss: first20 0.0310 -> last20 0.0260 (18960 items)
eval ep37: CER 0.000 exact 1.00 vram 4.41GB 17.4min
eval ep39: CER 0.000 exact 1.00 vram 4.41GB 18.3min
FULL eval: CER 0.0090 line-exact 0.9879 general-CE 4.129

== lw_k_final_0p6b_rag ==
method=layerwise schedule=summed lora=False lr=1e-05 epochs=40 ce=0.0/0.0 online=False
loss: first20 0.0738 -> last20 0.0125 (18960 items)
eval ep37: CER 0.042 exact 1.00 vram 9.92GB 34.2min
eval ep39: CER 0.000 exact 1.00 vram 9.92GB 36.3min
FULL eval: CER 0.0092 line-exact 0.9874 general-CE 3.862

== lw_k_final_1p7b_rag ==
method=layerwise schedule=summed lora=False lr=1e-05 epochs=40 ce=0.0/0.0 online=False
loss: first20 0.1603 -> last20 0.0242 (18960 items)
eval ep37: CER 0.090 exact 0.88 vram 27.91GB 51.4min
eval ep39: CER 0.101 exact 0.88 vram 27.91GB 53.7min
FULL eval: CER 0.0205 line-exact 0.9758 general-CE 3.603

== lw_k_final_k8_0p6b_rag ==
method=layerwise schedule=summed lora=False lr=1e-05 epochs=40 ce=0.0/0.0 online=False
loss: first20 0.0736 -> last20 0.0105 (18960 items)
eval ep37: CER 0.244 exact 0.75 vram 10.21GB 31.1min
eval ep39: CER 0.000 exact 1.00 vram 10.21GB 32.7min
FULL eval: CER 0.0151 line-exact 0.9860 general-CE 3.791

== lw_k_final_s43_0p6b_rag ==
method=layerwise schedule=summed lora=False lr=1e-05 epochs=40 ce=0.0/0.0 online=False
loss: first20 0.0754 -> last20 0.0119 (18960 items)
eval ep37: CER 0.000 exact 1.00 vram 9.92GB 30.3min
eval ep39: CER 0.042 exact 1.00 vram 9.92GB 31.9min
FULL eval: CER 0.0124 line-exact 0.9880 general-CE 3.918
```

Per-run details (6)

```
== lw_k_gptoss_rag ==
method=layerwise schedule=summed lora=True lr=0.0001 epochs=20 ce=0.0/0.0 online=True
loss: first20 0.2725 -> last20 0.1482 (6660 items)
eval ep17: CER 1.430 exact 0.00 vram 40.67GB 30.2min
eval ep19: CER 1.472 exact 0.00 vram 40.67GB 33.7min
FULL eval: CER 1.0000 line-exact 0.0000 general-CE 3.483

== lw_k_maieutic_0p6b_rag ==
method=layerwise schedule=summed lora=False lr=1e-05 epochs=40 ce=0.0/0.0 online=False
loss: first20 0.0736 -> last20 0.0125 (18960 items)
eval ep37: CER 0.070 exact 1.00 vram 8.66GB 29.6min
eval ep39: CER 0.000 exact 1.00 vram 8.66GB 31.2min
FULL eval: CER 0.0102 line-exact 0.9862 general-CE 4.458

== lw_k_tailonly_0p6b_rag ==
method=layerwise schedule=tail_only lora=False lr=1e-05 epochs=40 ce=0.0/0.0 online=False
loss: first20 0.0373 -> last20 0.0316 (13320 items)
eval ep37: CER 0.000 exact 1.00 vram 3.13GB 13.3min
eval ep39: CER 0.000 exact 1.00 vram 3.13GB 14.2min
FULL eval: CER 0.0081 line-exact 0.9902 general-CE 4.609

== lw_k_tailonly_anchor_0p6b_rag ==
method=layerwise schedule=tail_only lora=False lr=1e-05 epochs=40 ce=0.0/0.0 online=False
loss: first20 0.0373 -> last20 0.0323 (13320 items)
eval ep37: CER 0.000 exact 1.00 vram 3.13GB 10.0min
eval ep39: CER 0.000 exact 1.00 vram 3.13GB 10.6min
FULL eval: CER 0.0106 line-exact 0.9876 general-CE 5.291

== lw_k_tailonly_anchorkl_0p6b_rag ==
method=layerwise schedule=tail_only lora=False lr=1e-05 epochs=40 ce=0.0/0.0 online=False
loss: first20 0.0374 -> last20 0.0322 (13320 items)
eval ep37: CER 0.000 exact 1.00 vram 4.41GB 13.3min
eval ep39: CER 0.000 exact 1.00 vram 4.41GB 14.0min
FULL eval: CER 0.0102 line-exact 0.9872 general-CE 3.970

== lw_k_v4strict_0p6b_rag ==
method=layerwise schedule=summed lora=False lr=1e-05 epochs=40 ce=0.0/0.0 online=False
loss: first20 0.0736 -> last20 0.0120 (18960 items)
eval ep37: CER 0.732 exact 0.00 vram 7.28GB 26.2min
eval ep39: CER 0.754 exact 0.00 vram 7.28GB 27.4min
```

Per-run details (7)

```
== lw_k_vocab_s43_0p6b_rag ==
method=layerwise schedule=summed lora=False lr=1e-05 epochs=40 ce=0.0/0.0 online=False
loss: first20 0.0745 -> last20 0.0140 (13320 items)
eval ep37: CER 0.000 exact 1.00 vram 8.66GB 35.4min
eval ep39: CER 0.000 exact 1.00 vram 8.66GB 37.1min
FULL eval: CER 0.0286 line-exact 0.9641 general-CE 4.234

== lw_lens_ce_0p6b_rag ==
method=layerwise schedule=summed lora=False lr=1e-05 epochs=40 ce=0.0/0.0 online=False
loss: first20 6.0093 -> last20 0.4134 (9120 items)
eval ep37: CER 0.812 exact 0.00 vram 7.95GB 64.4min
eval ep39: CER 0.831 exact 0.00 vram 7.95GB 67.2min
FULL eval: CER 0.7960 line-exact 0.0258 general-CE 6.044

== lw_lens_ce_deep_0p6b_rag ==
method=layerwise schedule=summed lora=False lr=1e-05 epochs=40 ce=0.0/0.0 online=False
loss: first20 1.3734 -> last20 0.0918 (9120 items)
eval ep37: CER 0.275 exact 0.62 vram 7.95GB 40.7min
eval ep39: CER 0.245 exact 0.74 vram 7.95GB 42.7min
FULL eval: CER 0.3074 line-exact 0.7720 general-CE 4.907

== lw_lens_ce_lora_0p6b_rag ==
method=layerwise schedule=summed lora=True lr=0.0001 epochs=40 ce=0.0/0.0 online=True
loss: first20 6.0315 -> last20 1.9305 (9120 items)
eval ep37: CER 0.780 exact 0.00 vram 1.79GB 59.7min
eval ep39: CER 0.874 exact 0.00 vram 1.79GB 62.1min
FULL eval: CER 0.8375 line-exact 0.0000 general-CE 9.527

== lw_lens_kl_0p6b_rag ==
method=layerwise schedule=summed lora=False lr=1e-05 epochs=40 ce=0.0/0.5 online=False
loss: first20 1.3176 -> last20 0.1719 (9120 items)
eval ep37: CER 0.813 exact 0.00 vram 8.26GB 74.5min
eval ep39: CER 0.931 exact 0.00 vram 8.26GB 76.5min
FULL eval: CER 0.7056 line-exact 0.1233 general-CE 4.281

== lw_lens_kl_deep_0p6b_rag ==
method=layerwise schedule=summed lora=False lr=1e-05 epochs=40 ce=0.0/0.5 online=False
loss: first20 0.8917 -> last20 0.1016 (9120 items)
eval ep37: CER 0.785 exact 0.03 vram 8.26GB 40.4min
eval ep39: CER 0.805 exact 0.00 vram 8.26GB 42.3min
FULL eval: CER 0.7252 line-exact 0.0921 general-CE 4.448
```

Per-run details (8)

```
== lw_seq_0p6b_rag ==
method=layerwise schedule=sequential lora=False lr=1e-05 epochs=10 ce=0.0/0.0 online=False
stage losses: L1:0.001 L8:0.013 L15:0.034 L22:0.125
eval ep21: CER 1.030 exact 0.00 vram 1.43GB 39.1min
eval ep28: CER 0.866 exact 0.00 vram 1.46GB 50.0min
FULL eval: CER 0.9282 line-exact 0.0000 general-CE 3.676

== lw_seq_1p7b_rag ==
method=layerwise schedule=sequential lora=False lr=1e-05 epochs=10 ce=0.0/0.0 online=False
stage losses: L1:0.001 L8:0.019 L15:0.069 L22:0.143
eval ep21: CER 0.912 exact 0.00 vram 4.07GB 39.6min
eval ep28: CER 0.938 exact 0.00 vram 4.16GB 51.4min
FULL eval: CER 0.9273 line-exact 0.0000 general-CE 3.337

== lw_seq_bf16_0p6b_rag ==
method=layerwise schedule=sequential lora=False lr=1e-05 epochs=10 ce=0.0/0.0 online=False
stage losses: L1:0.001 L8:0.013 L15:0.034 L22:0.125
eval ep21: CER 1.037 exact 0.00 vram 1.43GB 37.1min
eval ep28: CER 0.835 exact 0.00 vram 1.46GB 47.3min
FULL eval: CER 0.9257 line-exact 0.0000 general-CE 3.684

== lw_summed_0p6b_rag ==
method=layerwise schedule=summed lora=False lr=1e-05 epochs=10 ce=0.0/0.0 online=False
loss: first20 0.2485 -> last20 0.0650 (2280 items)
eval ep7: CER 0.903 exact 0.00 vram 7.23GB 4.7min
eval ep9: CER 0.879 exact 0.00 vram 7.23GB 6.1min
FULL eval: CER 0.9055 line-exact 0.0000 general-CE 3.858

== lw_summed_ce_0p6b_rag ==
method=layerwise schedule=summed lora=False lr=1e-05 epochs=20 ce=0.0/0.5 online=False
loss: first20 0.3224 -> last20 0.0868 (4560 items)
eval ep17: CER 0.937 exact 0.00 vram 7.95GB 16.3min
eval ep19: CER 0.794 exact 0.00 vram 7.95GB 18.1min
FULL eval: CER 1.0005 line-exact 0.0006 general-CE 3.807

== lw_summed_ce_e40_0p6b_rag ==
method=layerwise schedule=summed lora=False lr=1e-05 epochs=40 ce=0.0/0.5 online=False
loss: first20 0.3224 -> last20 0.0575 (9120 items)
eval ep37: CER 0.786 exact 0.00 vram 7.95GB 32.6min
eval ep39: CER 0.875 exact 0.00 vram 7.95GB 34.2min
FULL eval: CER 0.8819 line-exact 0.0125 general-CE 4.230
```

Per-run details (9)

```
== lw_summed_e40_0p6b_rag ==
method=layerwise schedule=summed lora=False lr=1e-05 epochs=40 ce=0.0/0.0 online=False
loss: first20 0.2485 -> last20 0.0437 (9120 items)
eval ep37: CER 0.811 exact 0.00 vram 7.23GB 29.9min
eval ep39: CER 0.768 exact 0.00 vram 7.23GB 31.3min
FULL eval: CER 0.9348 line-exact 0.0004 general-CE 4.233

== lw_summed_l2mse_e40_0p6b_rag ==
method=layerwise schedule=summed lora=False lr=1e-05 epochs=40 ce=0.0/0.0 online=False
loss: first20 0.0002 -> last20 0.0000 (9120 items)
eval ep37: CER 0.799 exact 0.00 vram 7.23GB 19.9min
eval ep39: CER 0.764 exact 0.00 vram 7.23GB 20.7min
FULL eval: CER 0.9348 line-exact 0.0000 general-CE 4.336

== lw_tail_ce_0p6b_rag ==
method=layerwise schedule=summed lora=False lr=1e-05 epochs=20 ce=0.0/0.0 online=False
loss: first20 0.2486 -> last20 0.0597 (4560 items)
eval ep17: CER 0.912 exact 0.00 vram 8.12GB 15.2min
eval ep19: CER 0.728 exact 0.04 vram 8.12GB 16.6min
FULL eval: CER 0.7916 line-exact 0.0500 general-CE 3.977

== lw_tail_ce_e40_0p6b_rag ==
method=layerwise schedule=summed lora=False lr=1e-05 epochs=40 ce=0.0/0.0 online=False
loss: first20 0.2486 -> last20 0.0462 (9120 items)
eval ep37: CER 0.402 exact 0.50 vram 8.12GB 25.1min
eval ep39: CER 0.253 exact 0.73 vram 8.12GB 26.8min
FULL eval: CER 0.1423 line-exact 0.8230 general-CE 4.226

== lw_tail_ce_e40_s2_0p6b_rag ==
method=layerwise schedule=summed lora=False lr=1e-05 epochs=40 ce=0.0/0.0 online=False
loss: first20 0.2583 -> last20 0.0476 (9120 items)
eval ep37: CER 0.477 exact 0.43 vram 8.12GB 25.2min
eval ep39: CER 0.351 exact 0.57 vram 8.12GB 26.4min
FULL eval: CER 0.1926 line-exact 0.7485 general-CE 4.305

== lw_tail_ce_e40_v2_0p6b_rag ==
method=layerwise schedule=summed lora=False lr=1e-05 epochs=40 ce=0.0/0.0 online=False
loss: first20 0.2615 -> last20 0.0418 (13320 items)
eval ep37: CER 0.167 exact 0.70 vram 8.63GB 21.0min
eval ep39: CER 0.340 exact 0.88 vram 8.63GB 22.0min
FULL eval: CER 0.1116 line-exact 0.9051 general-CE 4.584
```


Per-run details (10)

```
== lw_tail_ce_k2_e40_0p6b_rag ==
method=layerwise schedule=summed lora=False lr=1e-05 epochs=40 ce=0.0/0.0 online=False
loss: first20 0.2486 -> last20 0.0467 (9120 items)
eval ep37: CER 0.798 exact 0.00 vram 8.0GB 13.9min
eval ep39: CER 0.782 exact 0.00 vram 8.0GB 14.6min
FULL eval: CER 0.5345 line-exact 0.3091 general-CE 4.290

== lw_tail_ce_lora_0p6b_rag ==
method=layerwise schedule=summed lora=True lr=0.0001 epochs=40 ce=0.0/0.0 online=True
loss: first20 0.2504 -> last20 0.0670 (9120 items)
eval ep37: CER 0.752 exact 0.00 vram 1.86GB 31.3min
eval ep39: CER 0.782 exact 0.00 vram 1.86GB 33.0min
FULL eval: CER 0.7834 line-exact 0.0183 general-CE 4.377

== lw_tc_lora_14b_rag ==
method=layerwise schedule=teacher_censored lora=True lr=0.0001 epochs=20 ce=0.0/0.0 online=True
loss: first20 0.0079 -> last20 0.0025 (4560 items)
eval ep17: CER 0.886 exact 0.00 vram 29.04GB 23.3min
eval ep19: CER 0.897 exact 0.00 vram 29.04GB 25.4min
FULL eval: CER 0.8657 line-exact 0.0000 general-CE 1.842

== lw_tc_lora_e40_0p6b_rag ==
method=layerwise schedule=teacher_censored lora=True lr=1e-05 epochs=40 ce=0.0/0.0 online=True
loss: first20 0.0220 -> last20 0.0100 (9120 items)
eval ep37: CER 0.872 exact 0.00 vram 1.37GB 35.9min
eval ep39: CER 0.923 exact 0.00 vram 1.37GB 37.7min
FULL eval: CER 0.9019 line-exact 0.0000 general-CE 3.646
```